

Photosynthesis Overview

Photosynthesis is the process by which plants, algae, and some bacteria convert light energy into chemical energy.

The overall reaction can be summarized as: carbon dioxide plus water, in the presence of light, yields glucose plus oxygen.

The Light-Dependent Reactions

The light-dependent reactions take place in the thylakoid membrane of the chloroplast. Chlorophyll absorbs photons of light energy.

The electron transport chain depends on chlorophyll to capture light energy. ATP and NADPH produced here are used in the Calvin cycle.

The Calvin Cycle

The Calvin cycle, also called the light-independent reactions, occurs in the stroma of the chloroplast. It uses ATP

The Calvin cycle depends on ATP and NADPH from the light-dependent reactions. RuBisCO is essential to the c